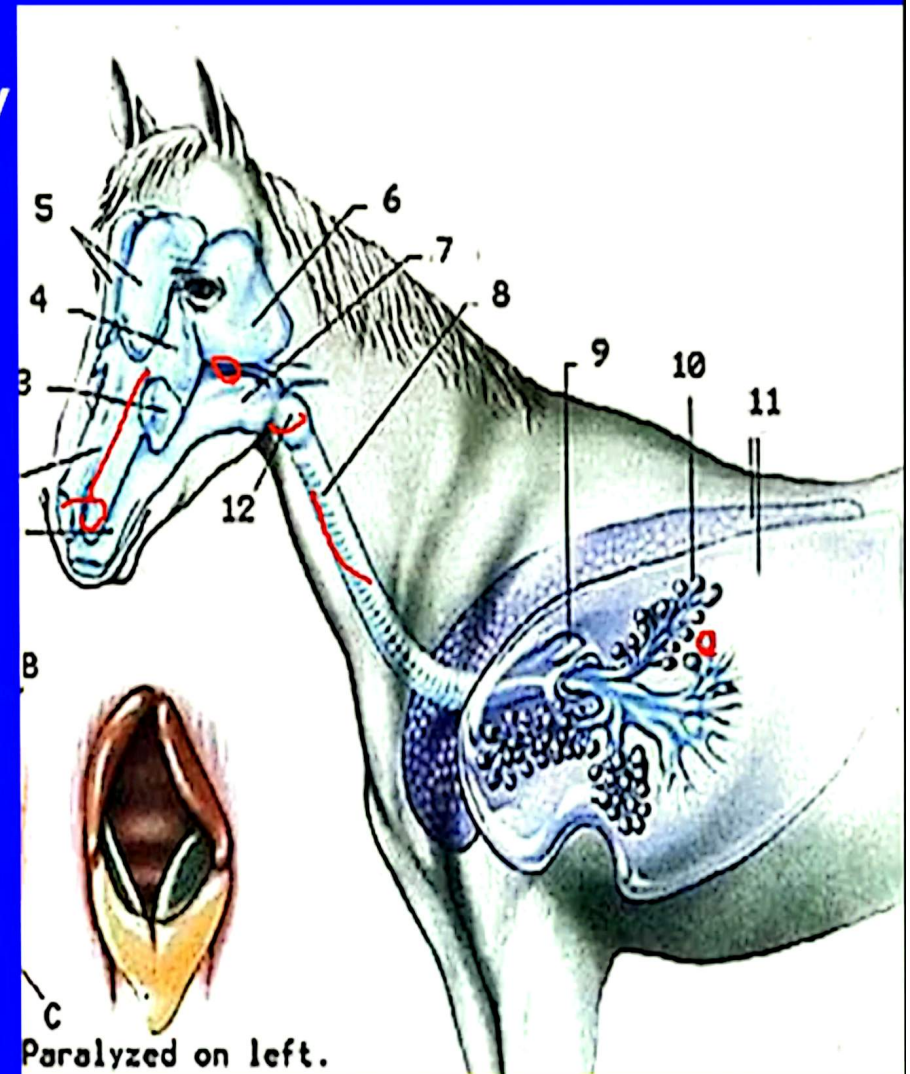


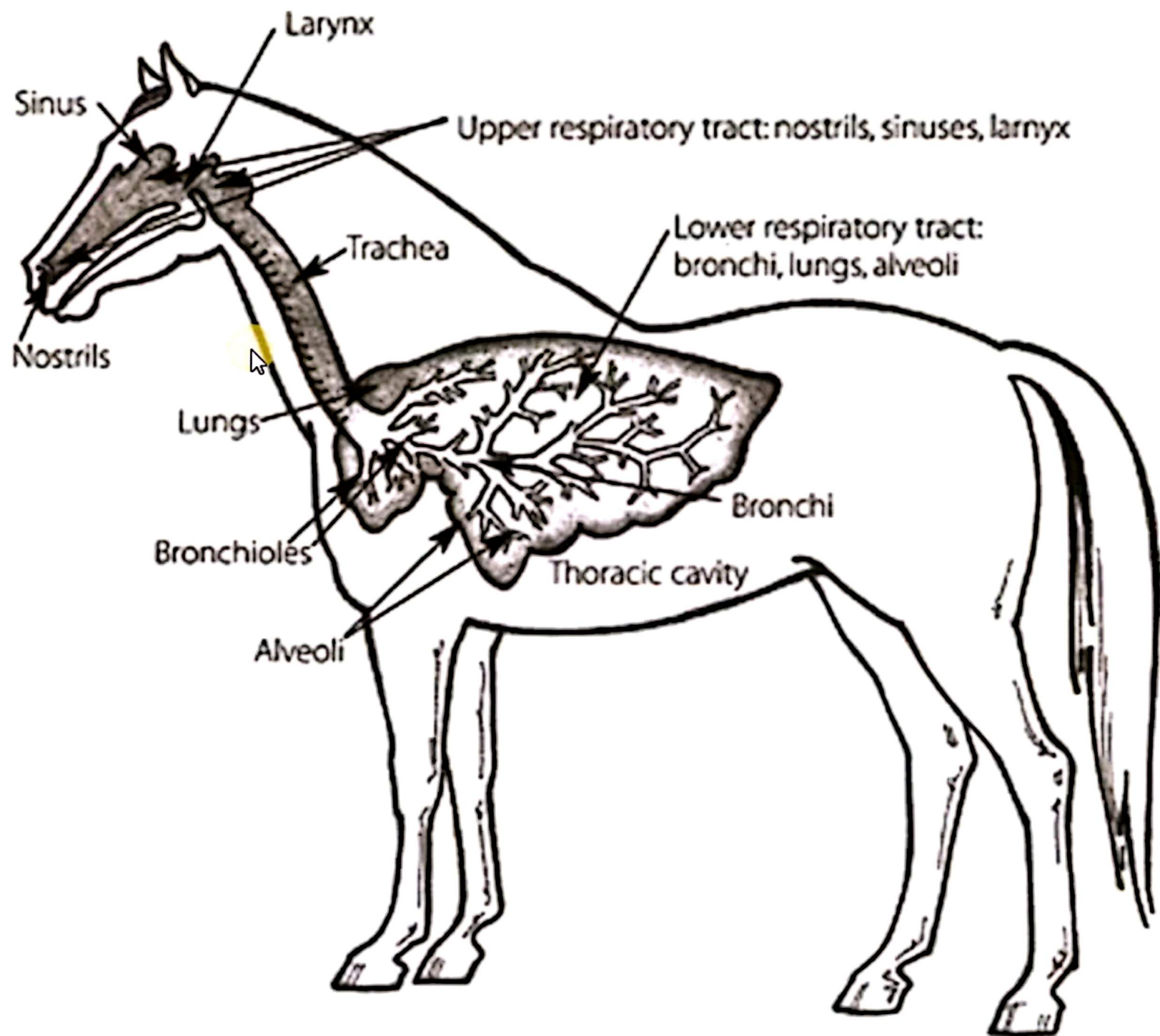
Parts of respiratory system

1-Conducting portion
external nose, nasal cavity
nasopharynx, larynx
, trachea and bronchi.

2- respiratory portion ✓
respiratory bronchiole,
alveolar sac.....

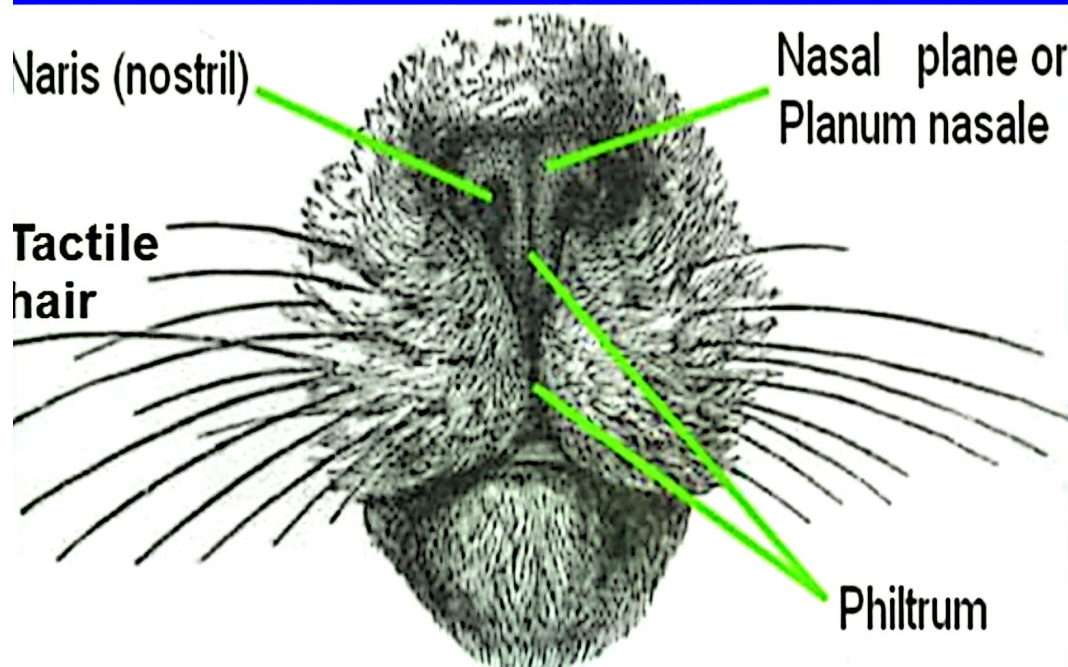
3-pumping portion
Diaphragm , muscles



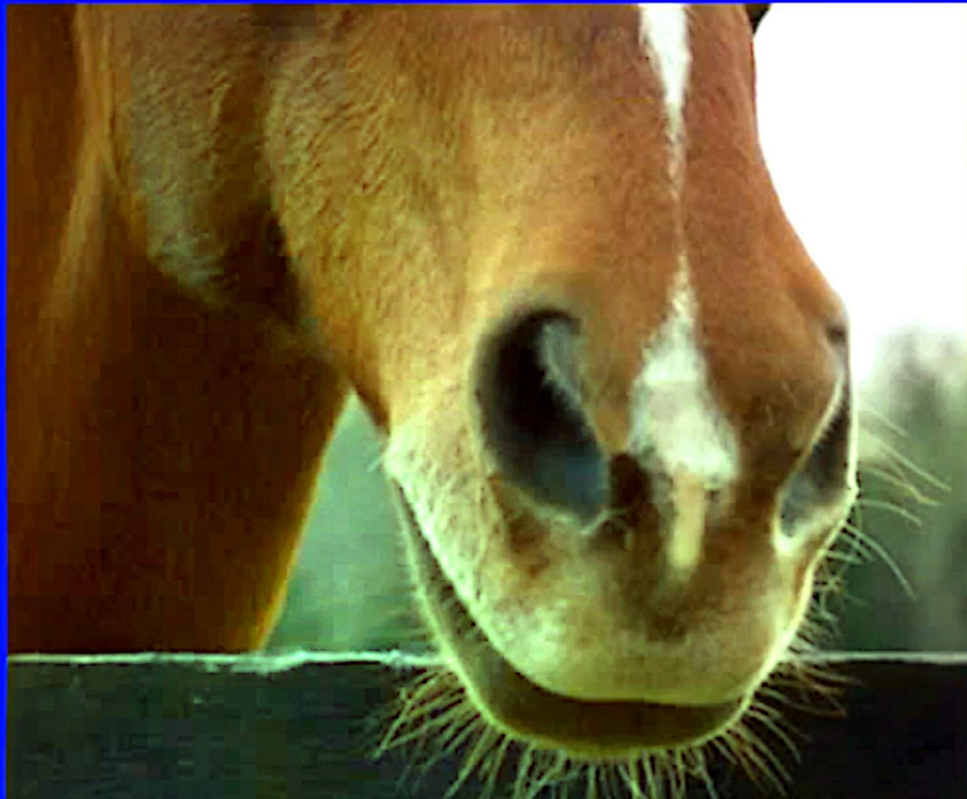


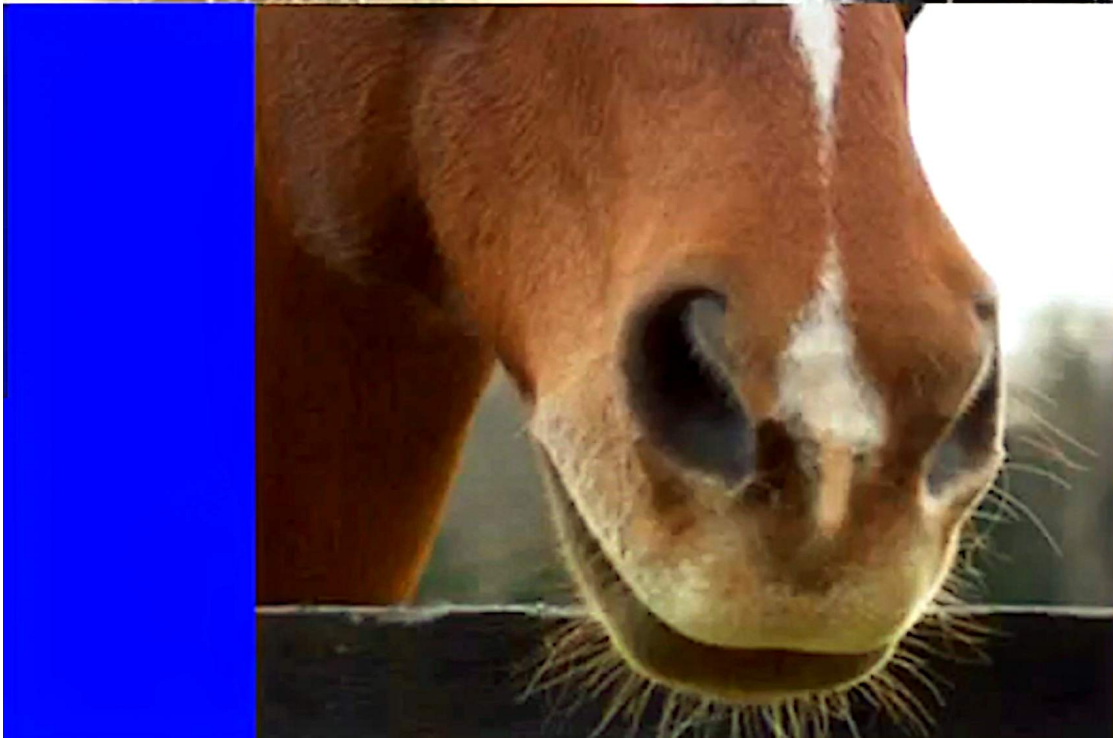
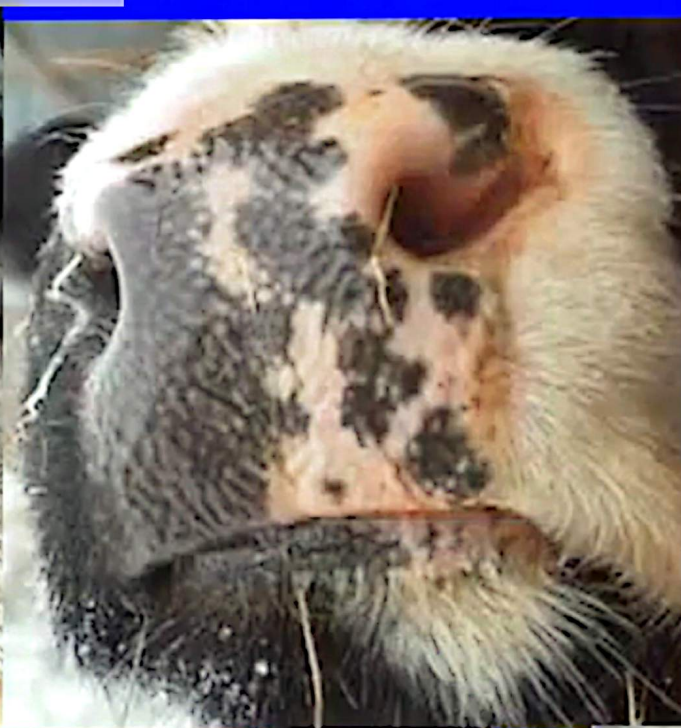
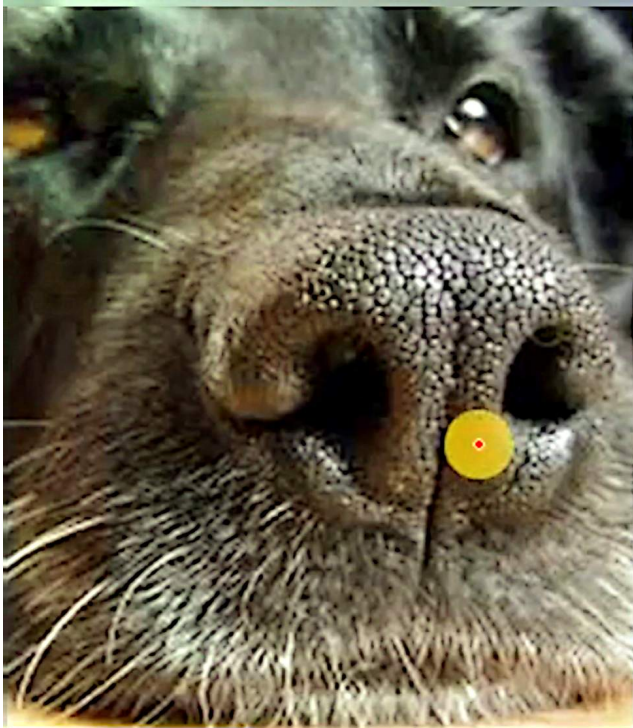
Nose

- ❑ In domestic animals embedded in the face. Projected in human and apex of nose protruded from the face in dog
- ❑ **Nose = external nose (nostrils & surrounding skin) + nasal cavity**



- The integument around the nostrils is modified hairless area **except at horse.**,
- The modified area is know as
 - ❑ the nasal plane (carnivores, small ruminants),
 - ❑ nasolabial plane (cattle)and
- The plate is kept moist in cattle,, and dogs;
- Tactile hair.







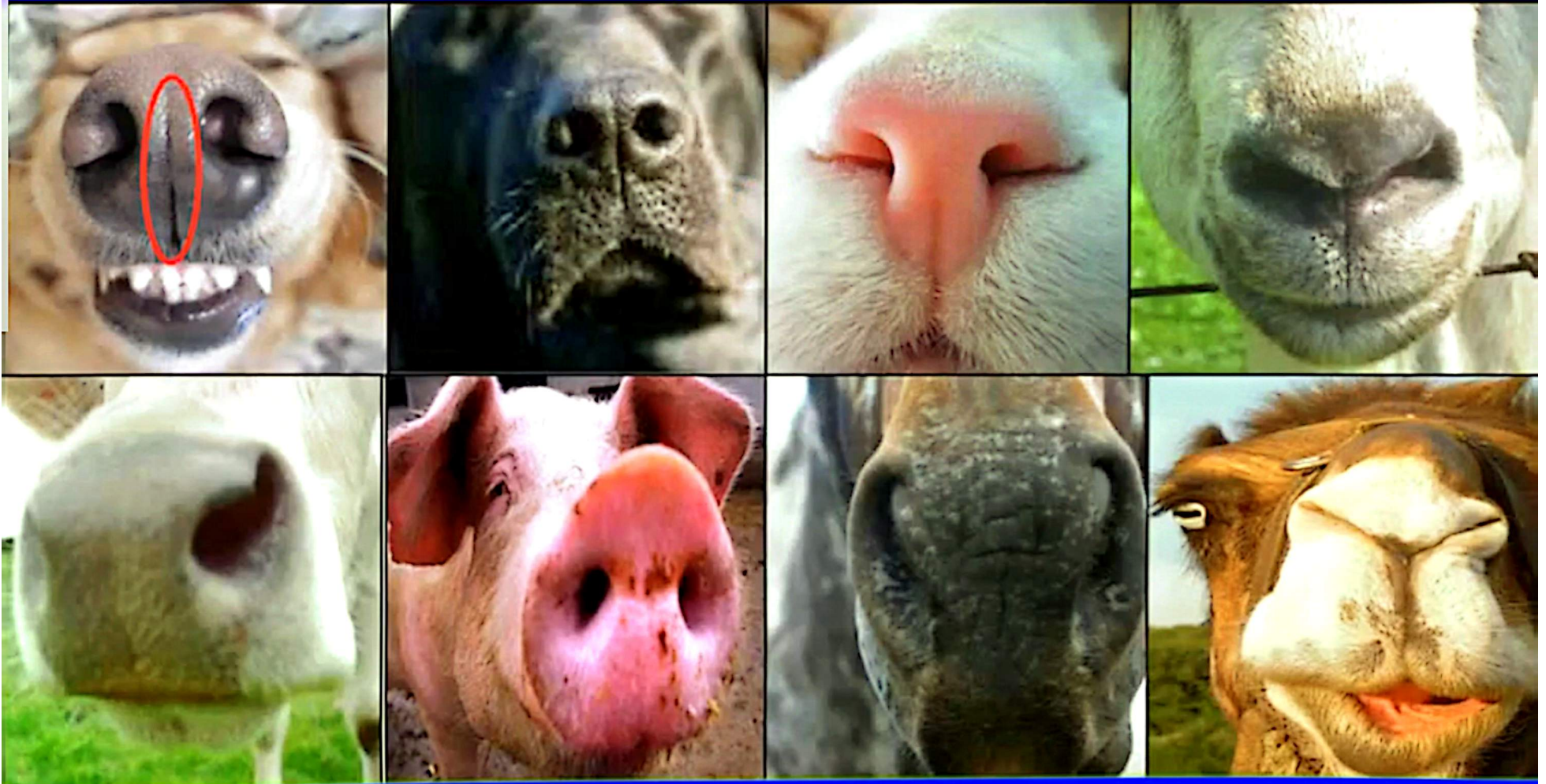
Planum Nasolabiale (Nasolabial plane):

The area of modified skin **around** and **between** the **Nostrils** and the skin of the **upper lip**.

It is smooth and devoid of hair except along lateral margin of the plane which has **Tactile hairs**.

Philtrum: is groove divided this area with upper lip into two area.

- In camel, dog and sheep is deep and extends to nostrils
- In the, ox and horse, it is shallow or absent.



Wall of nose

1- skin

2- supporting layer

- cartilage rostrally
- bone caudally

3- mucous membrane

□ The size and shape of nostrils, highly variable among domestic farm animals, are largely dictated by the nasal cartilages

- Shape of the nostrils (nasal opening):
 - oval or comma shape in dog., cat and ox
 - round in pig
 - Slit like in camel and sheep and goat.
- In horse: are **semilunar** in outline during normal breath and it is **circular** where respiratory increasing.

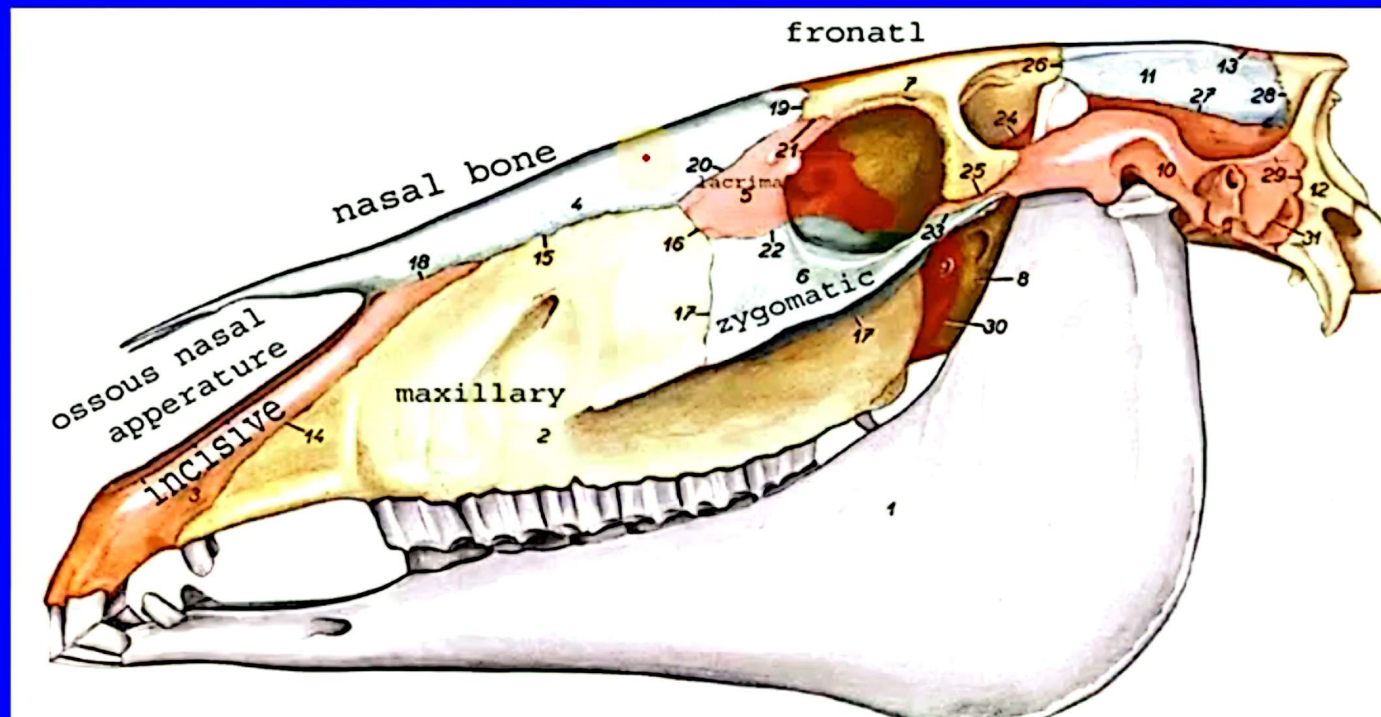


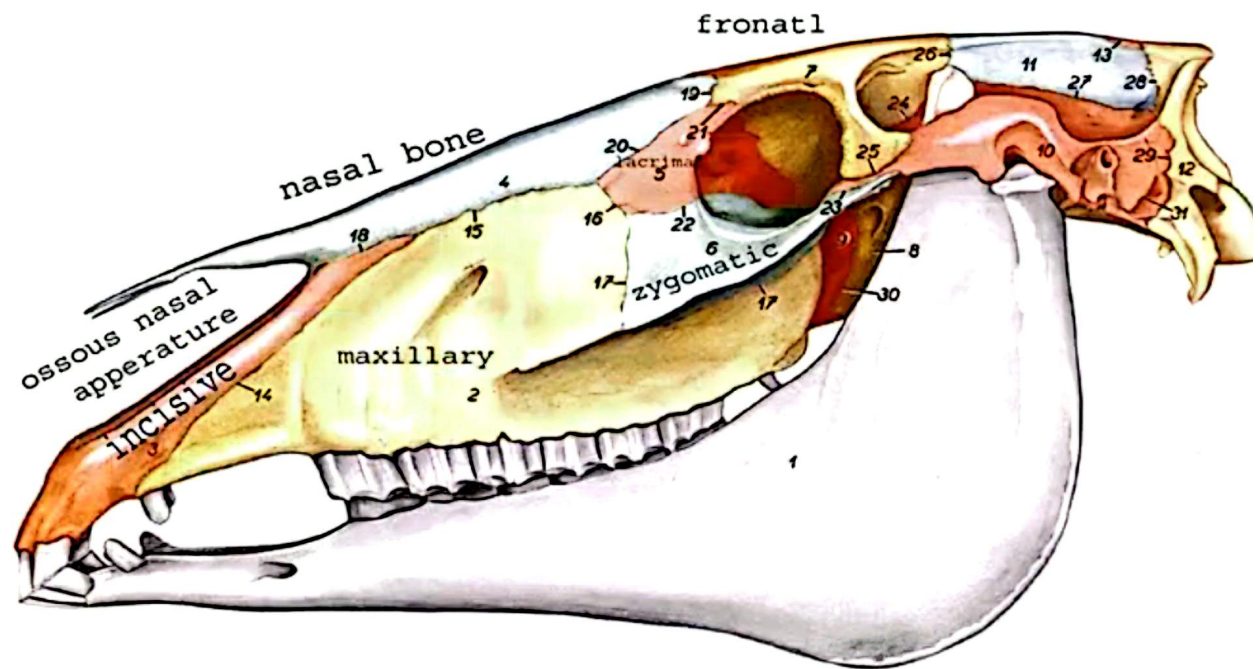
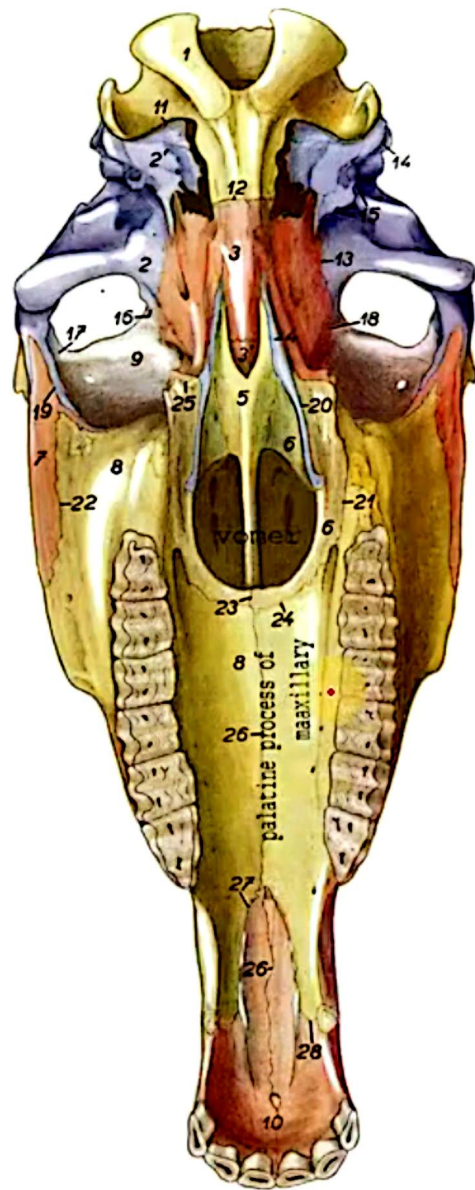
Bony Boundary

Dorsally : nasal , frontal bone

Laterally: incisive, maxillary, lacrimal and zygomatic bone

Ventrally :palatine process of incisive & maxillary , palatine bone





Cartilages of nostrilis:

1- Alar cartilage:

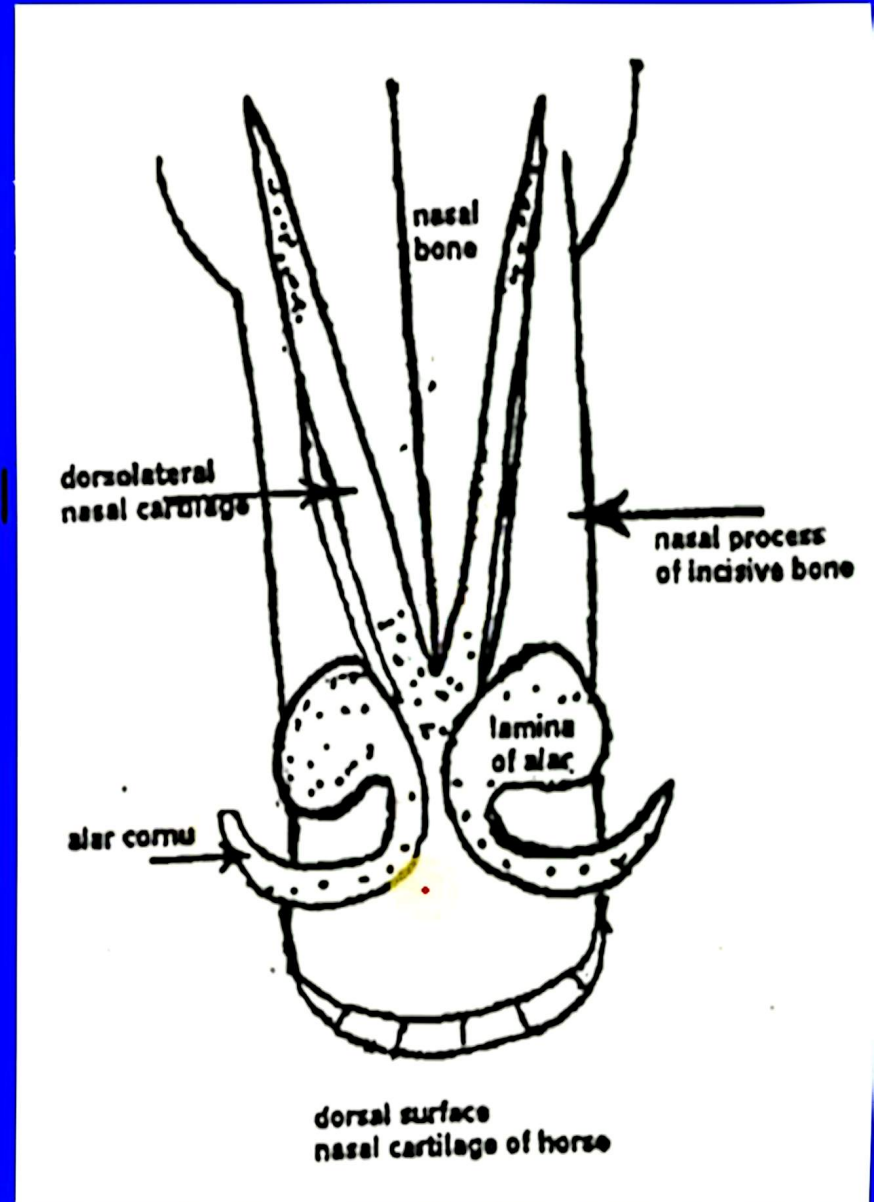
- only in horse ,
- comma shap ,(lamina& cornu
- attached to the rostral border of nasal septum,

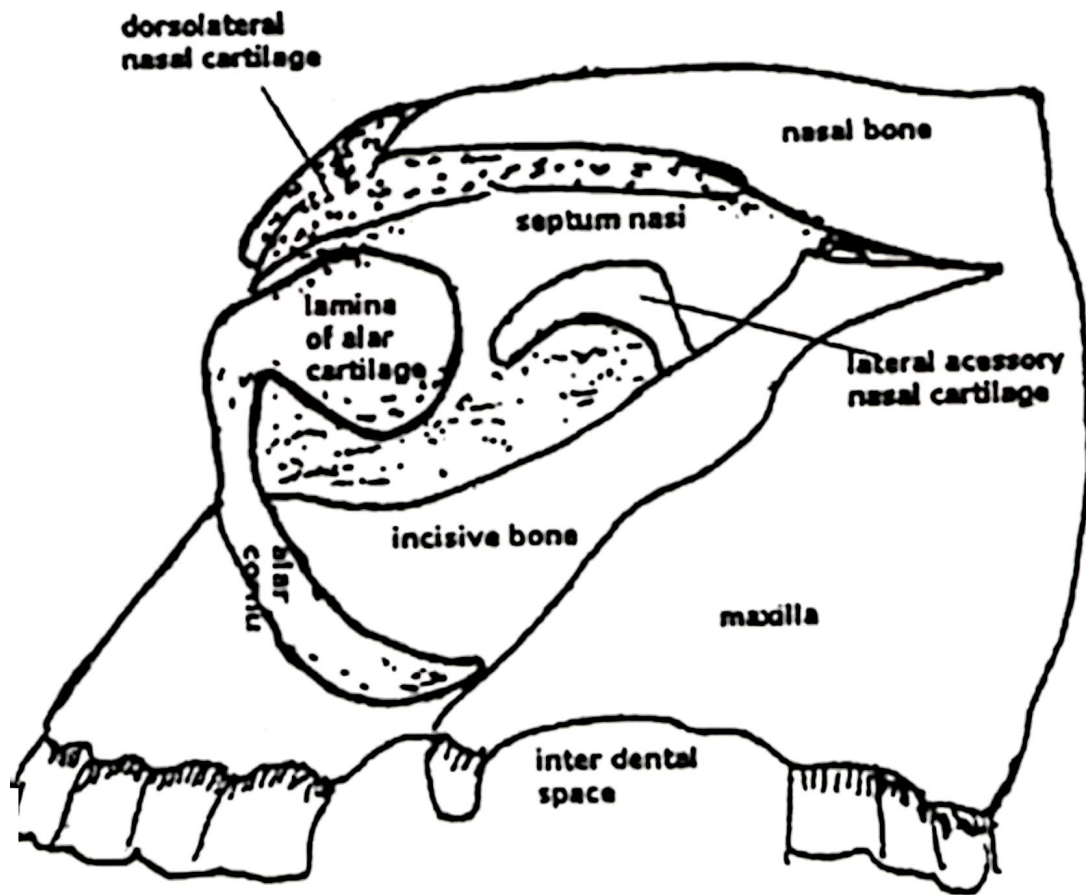
2-dorsal lateral & ventral lateral nasal cartilages

□ D. L N. present at nasal process .

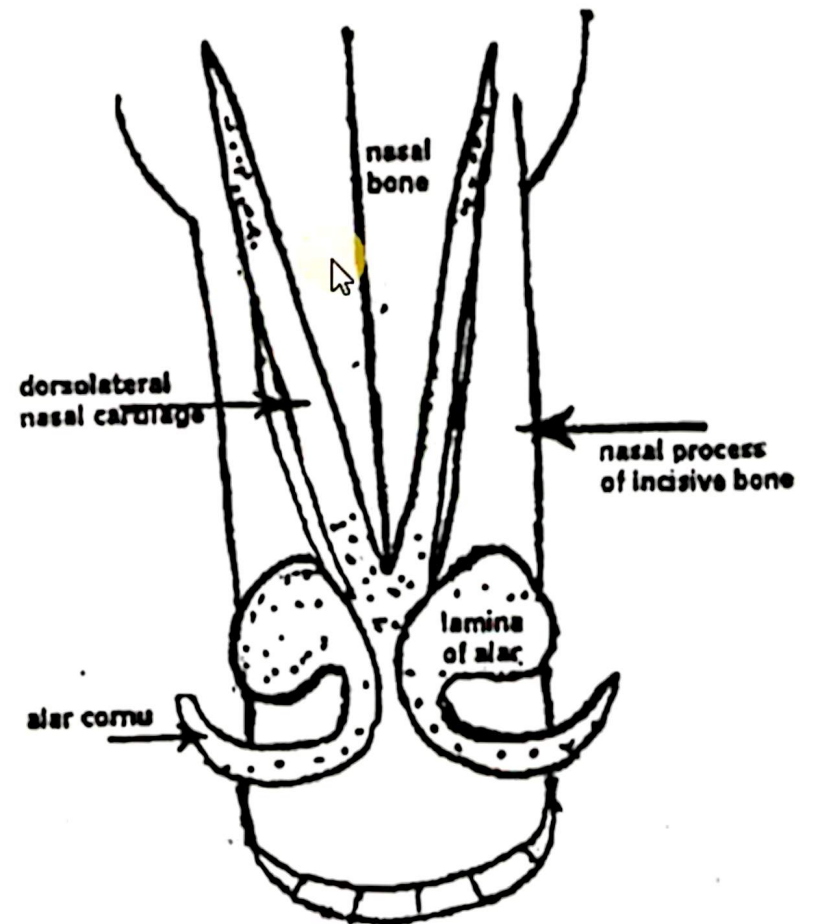
□ V.I.C. in palatine fissure
in horse may be absent

3-medial accessory nasal cartilage
present inside the alar fold
small in all species except horse
Large S shape





lateral view of nasal cartilages of horse



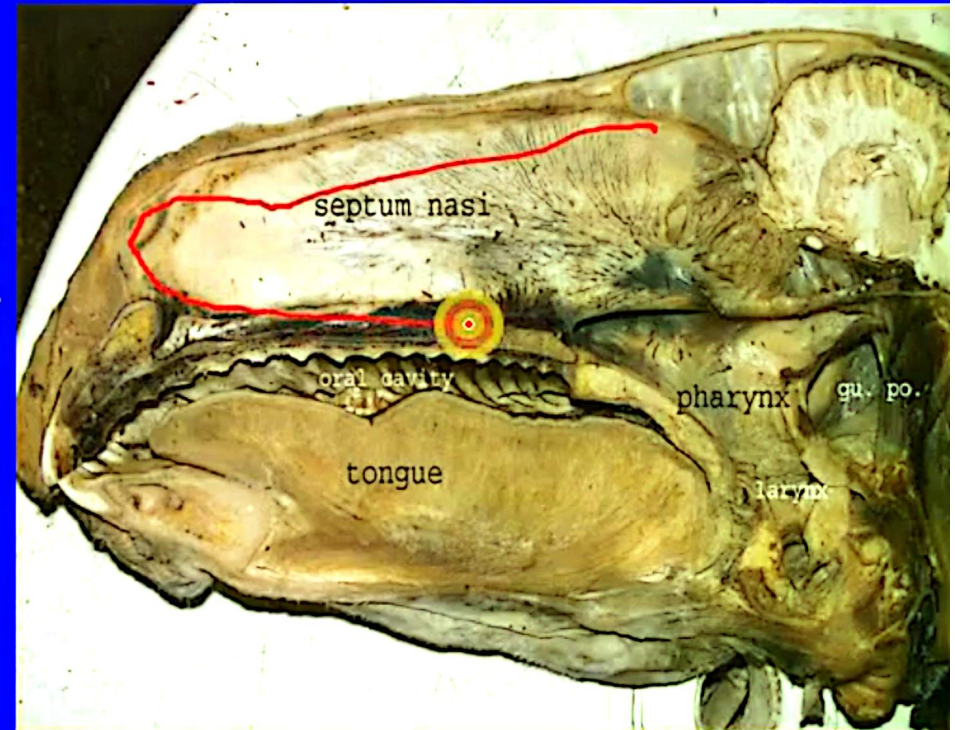
dorsal surface nasal cartilage of horse

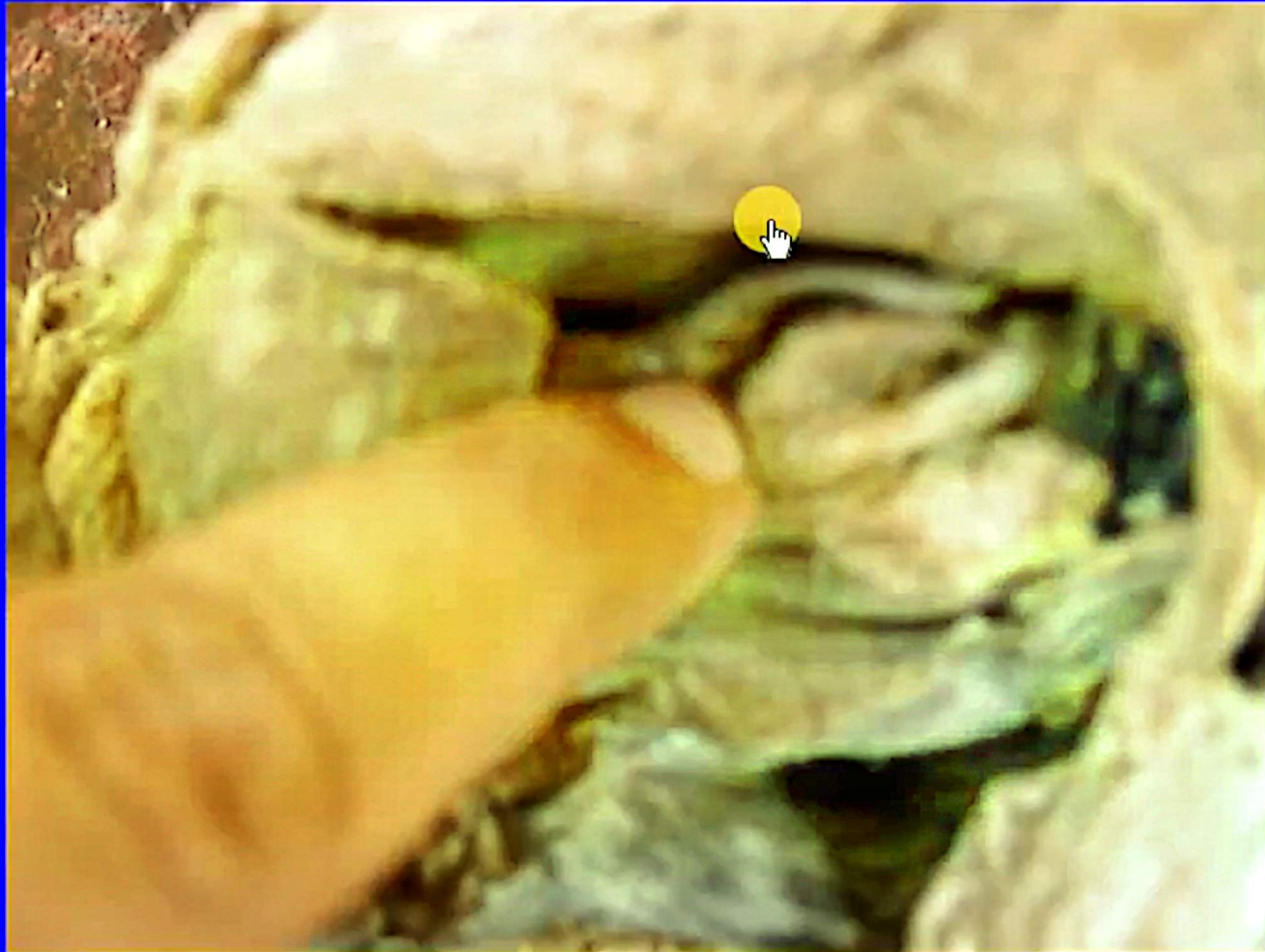
4-lateral accessory nasal cartilage

❑ Absent in horse

5- septum nasi

It is partition bet. 2 nostrils and divided the nasal cavity into right & left halves ,
Its caudal part is ossous & rostral is cartiaginous





Choana Nasopharynx

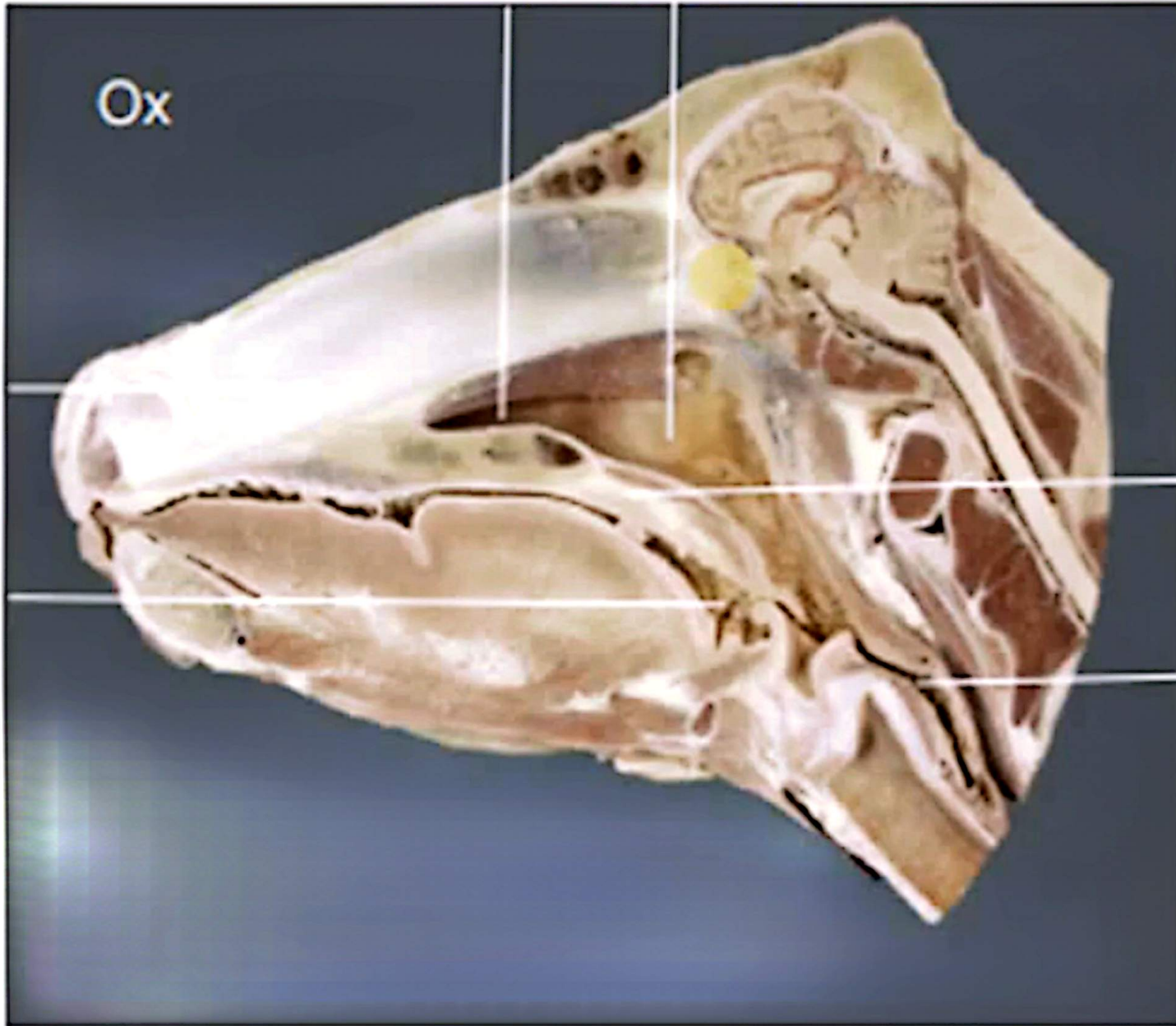
Ox

Nasal septum
(cartilaginous
part)

Oropharynx

Soft palate

Laryngopharynx



Nostrilis

Shape :

Dilated..... round

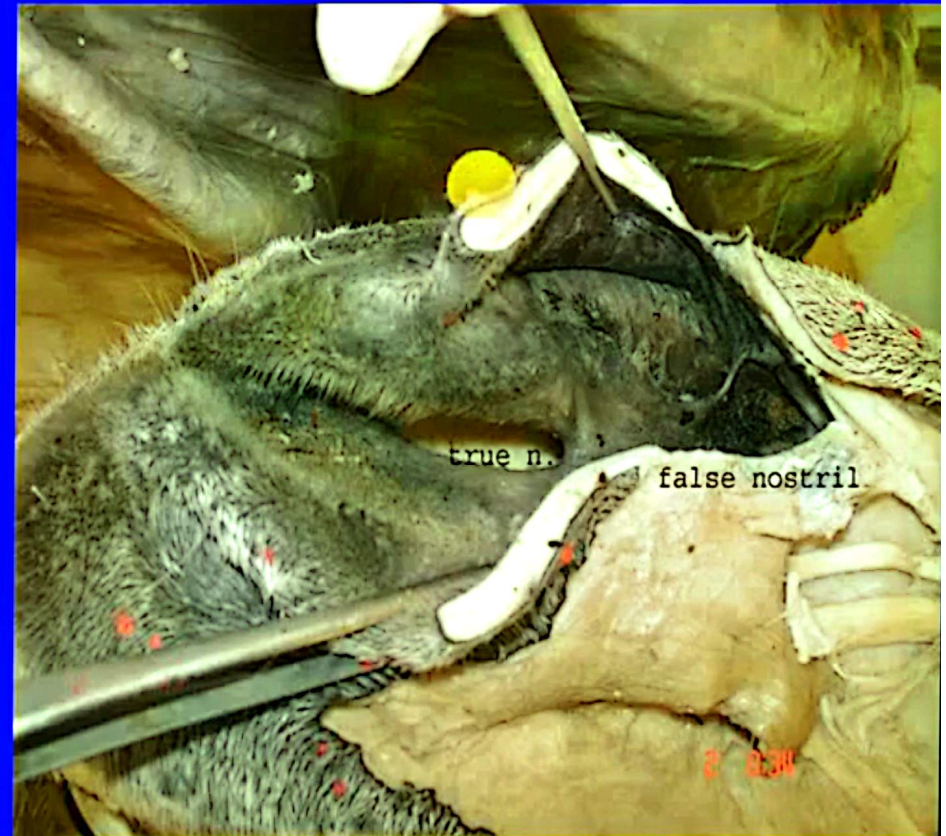
un dilated ...semilunar

It divided into two parts

1- false n.(nasal diverticulum)

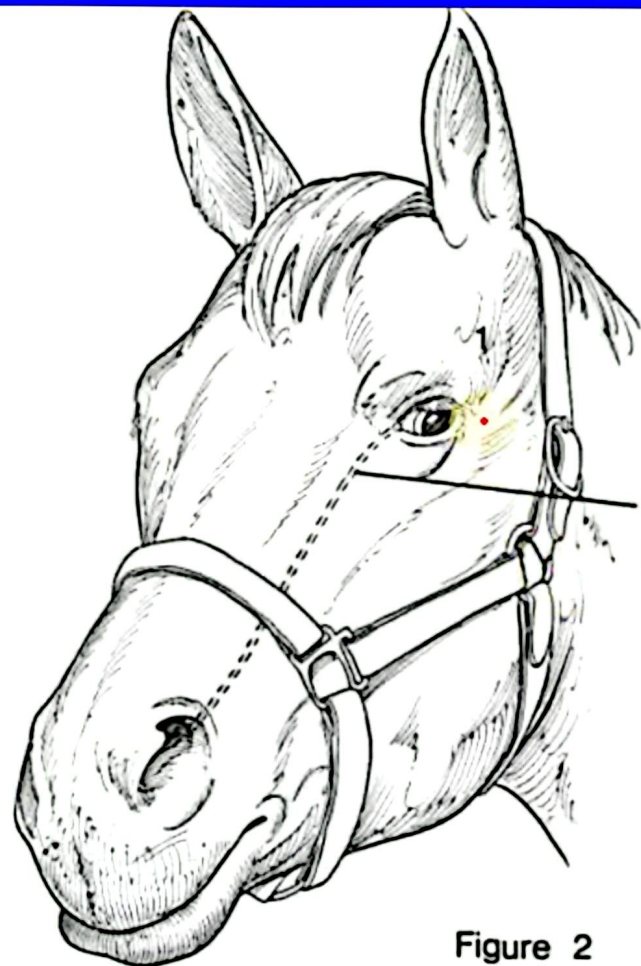
dorsal passage and form the blind pouch & is probably a construct that aids in passive dilation of the nostrils during vigorous ventilation.

2- true n. lead to nasal cavity





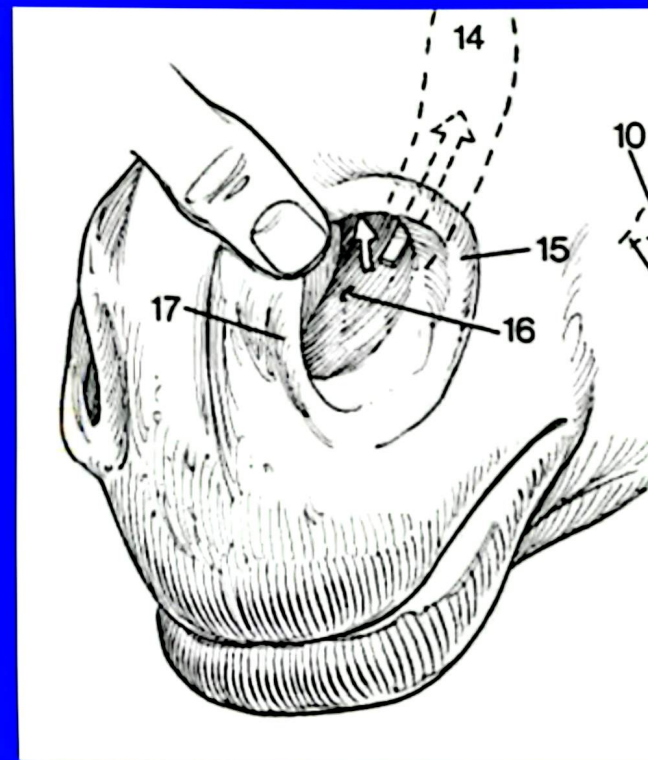
1



Course of
nasolacrimal duct

Figure 2

1



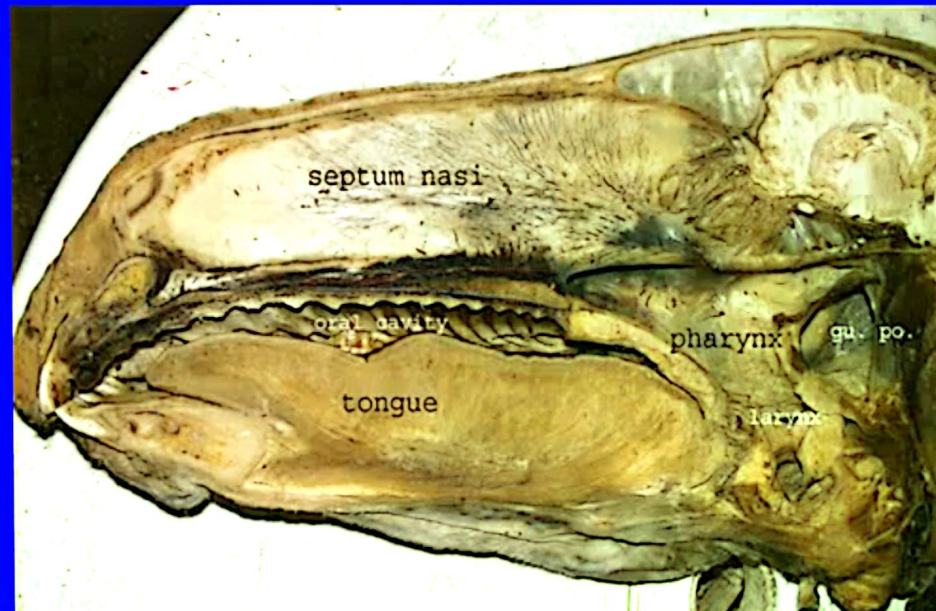
Nasal cavity

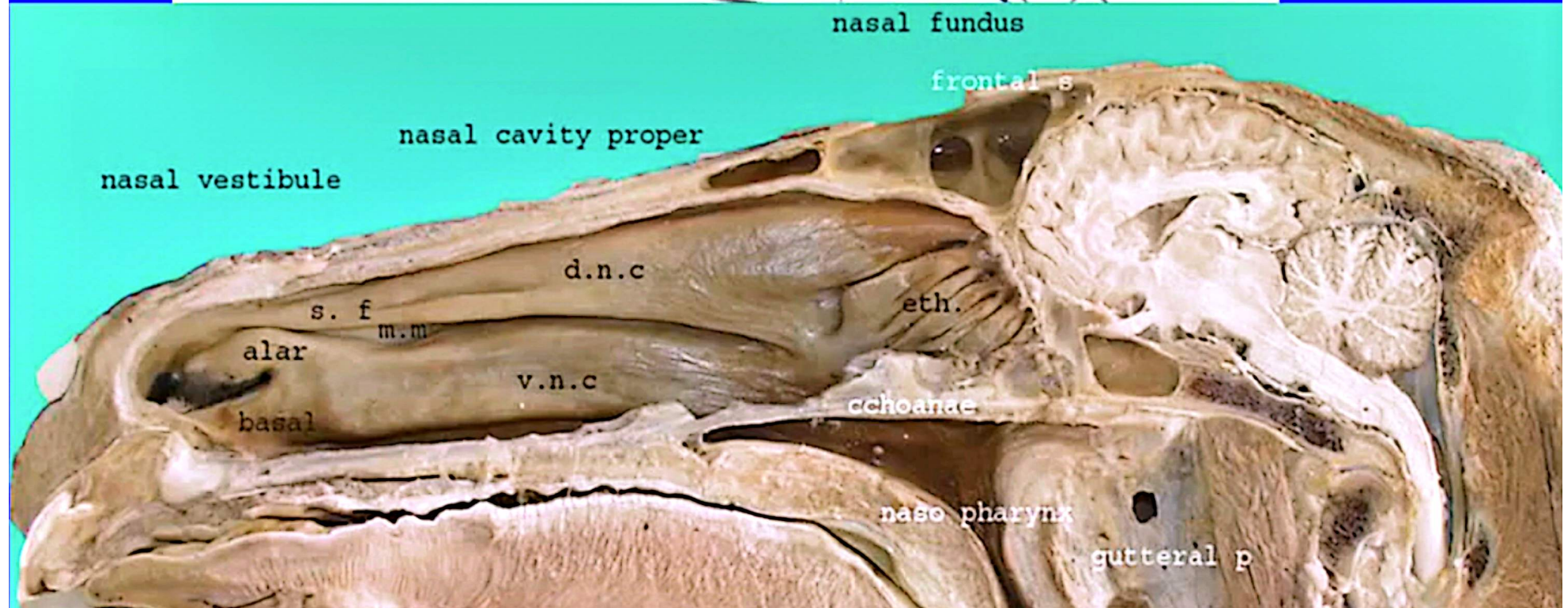
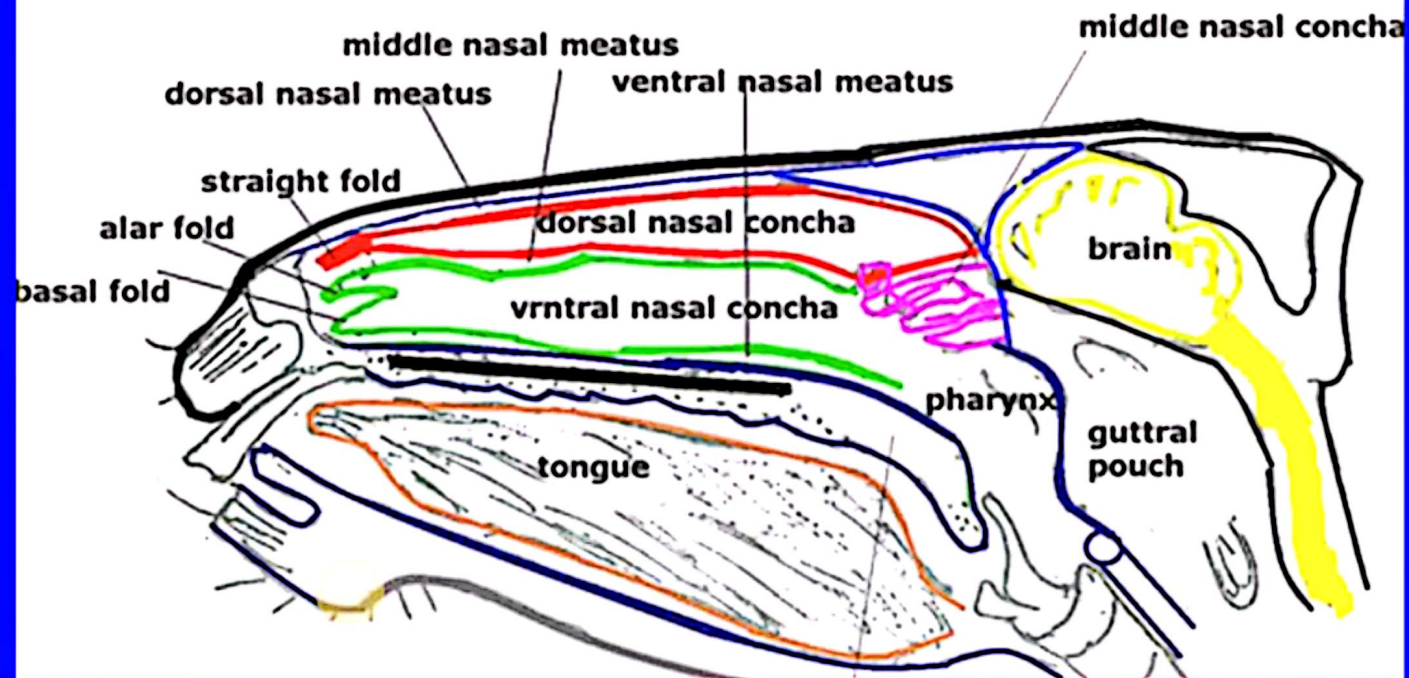
Def:

It is a part of conducting system, open externally the nostrils and communicates caudally with pharynx through the caudal narse or choanae and communicated by the paranasal sinus

Boundaries:

Dorsal Ventral Lateral Medially : septum nasi





Parts of nasal cavity

1- nasal vestibule 2- nasal cavity proper 3- nasal funds

1-nasal vestibule

- ❑ rostral part of N.C. & Lined by cutaneous m.m
- ❑ 3 folds 1. stright fold 2. alar fold 3.basal fold
- ❑ Contained nasolacrima opening

2-nasal cavity proper

- ❑ It is largest and middle part of nasal cavity & lined by respiratory m m.
- ❑ Consisted of dorsal nasal concha ventral nasal concha
 - dorsal nasal meatus Olfactory (nasal fundus)
 - middle nasal meatus.... Sinus meatus (naso maxillary opening
 - ventral nasal meatusrespiratory (nasopharynx)
 - Commen nasal meatus

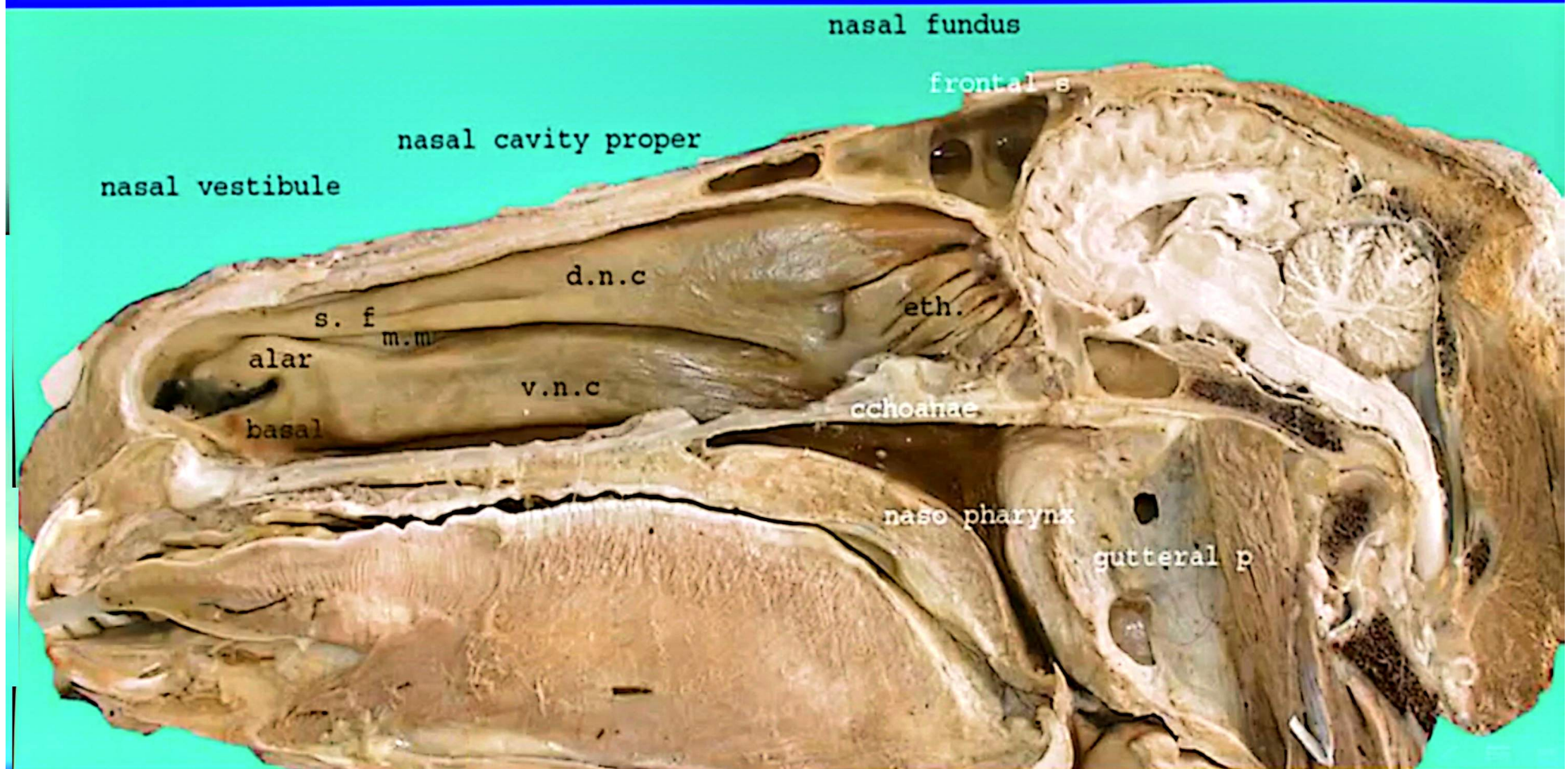


3-nasal fundus

It is caudal part of nasal cavity & lined by olfactory m.m

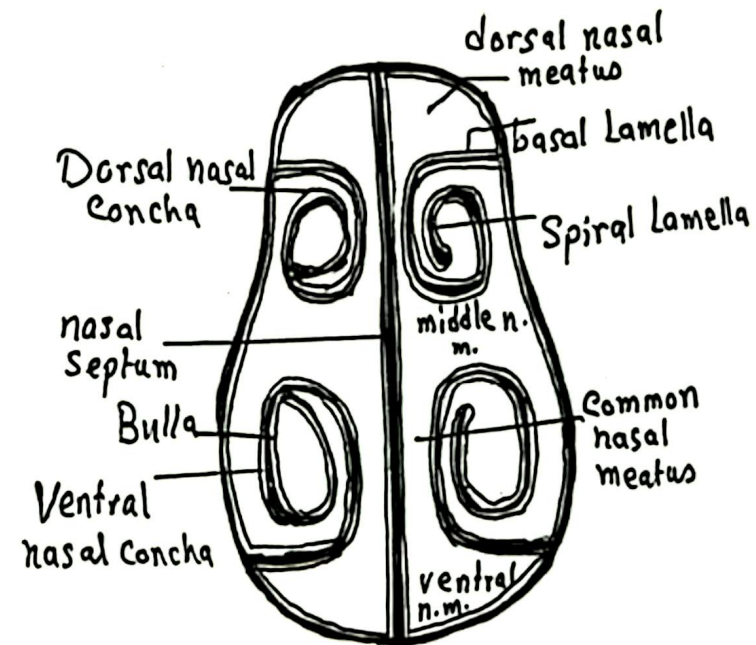
It consisted of

- 1- ethmoidal concha (olfactory part)
- 2- choanae(caudal nars) 2 elliptical orifice w.

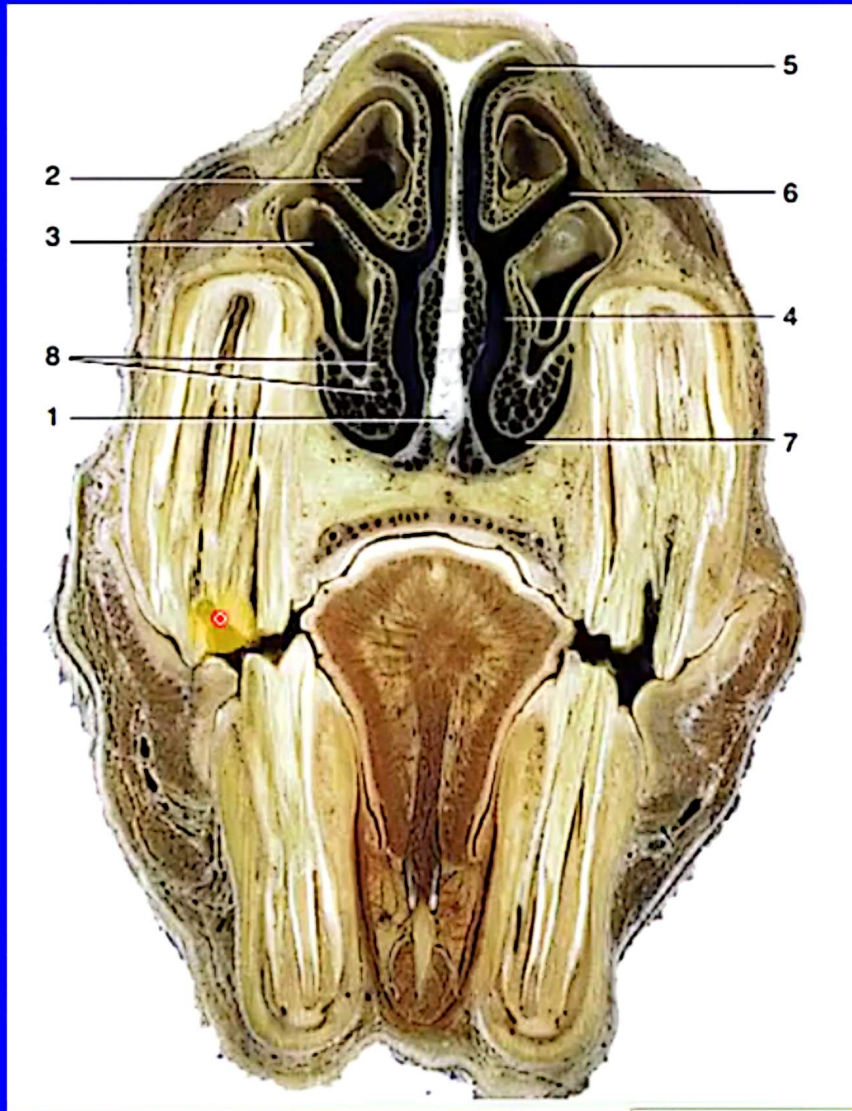


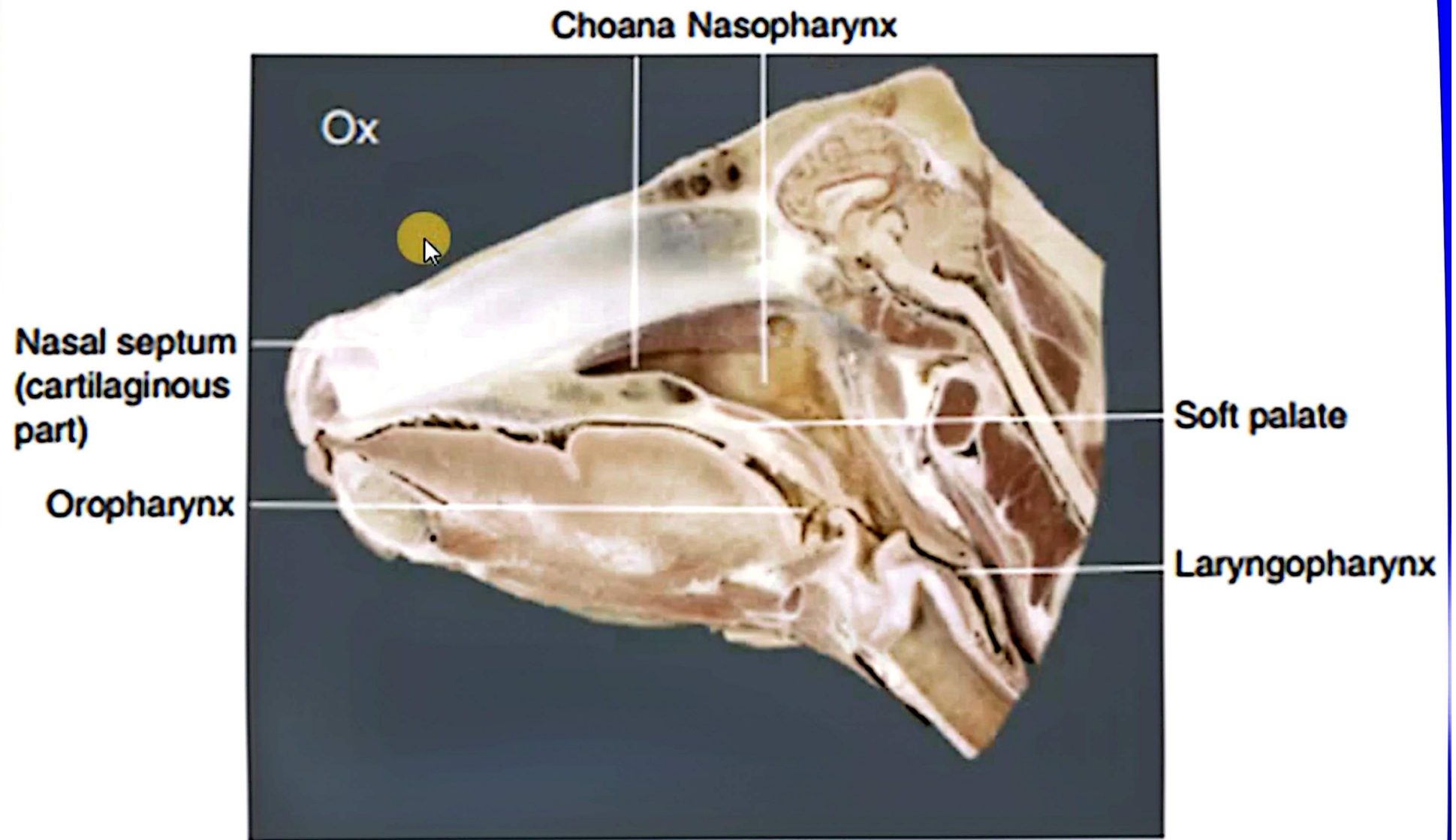
Nasal chonchae:

- ❖ thin osseous scrolls which consist of 2 part ,
- ❖ basal lamina w. fixed on the dorsal and ventral nasal crest
- ❖ then made spiral lamina , roll up to form bulla w . Divided by several septa to cellullae

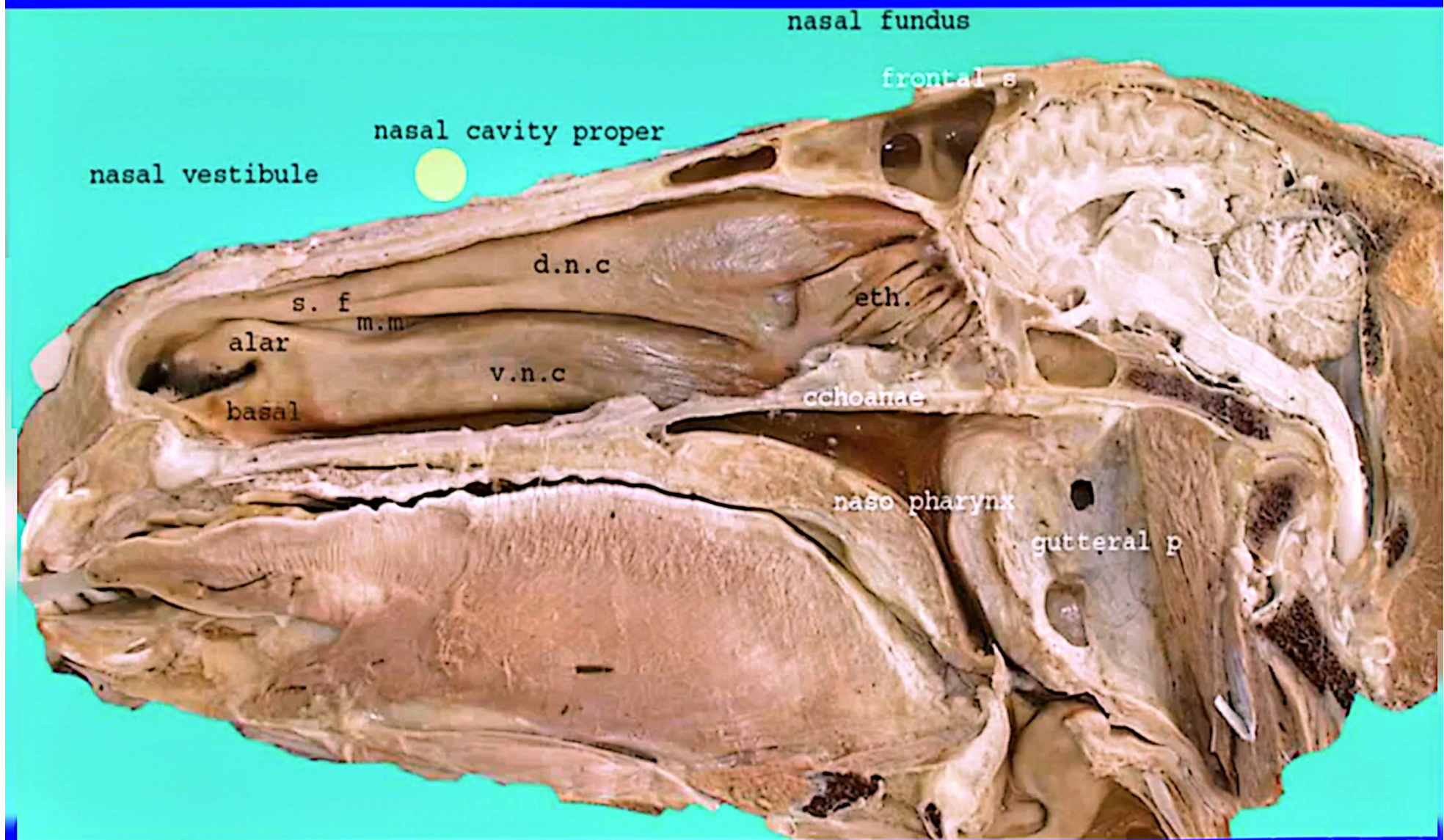


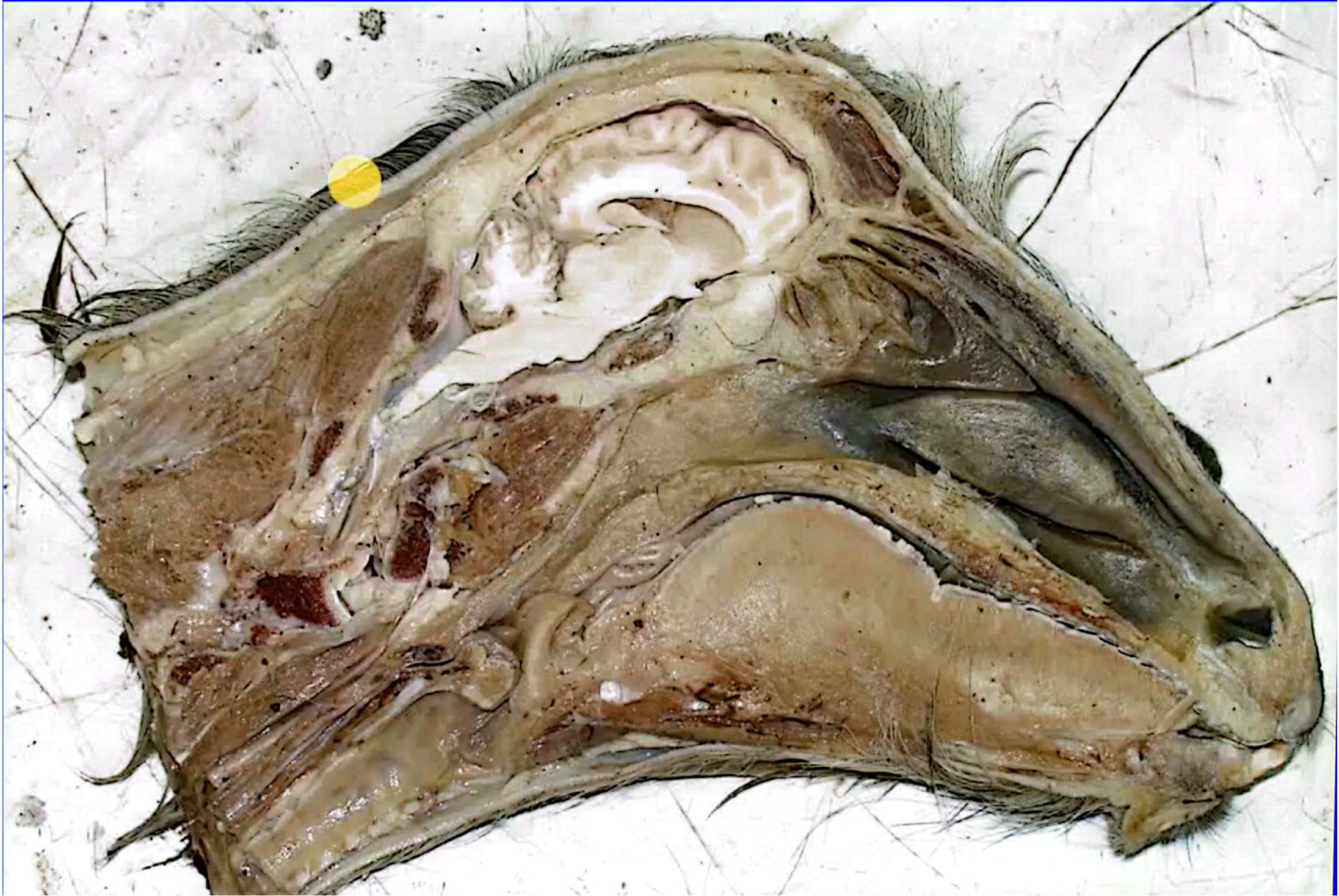
Cross section of nasal cavity of the horse. The thin lines represent the nasal mucosa.





Sagittal section in head





Vemronasal organ(organs of jacobson):

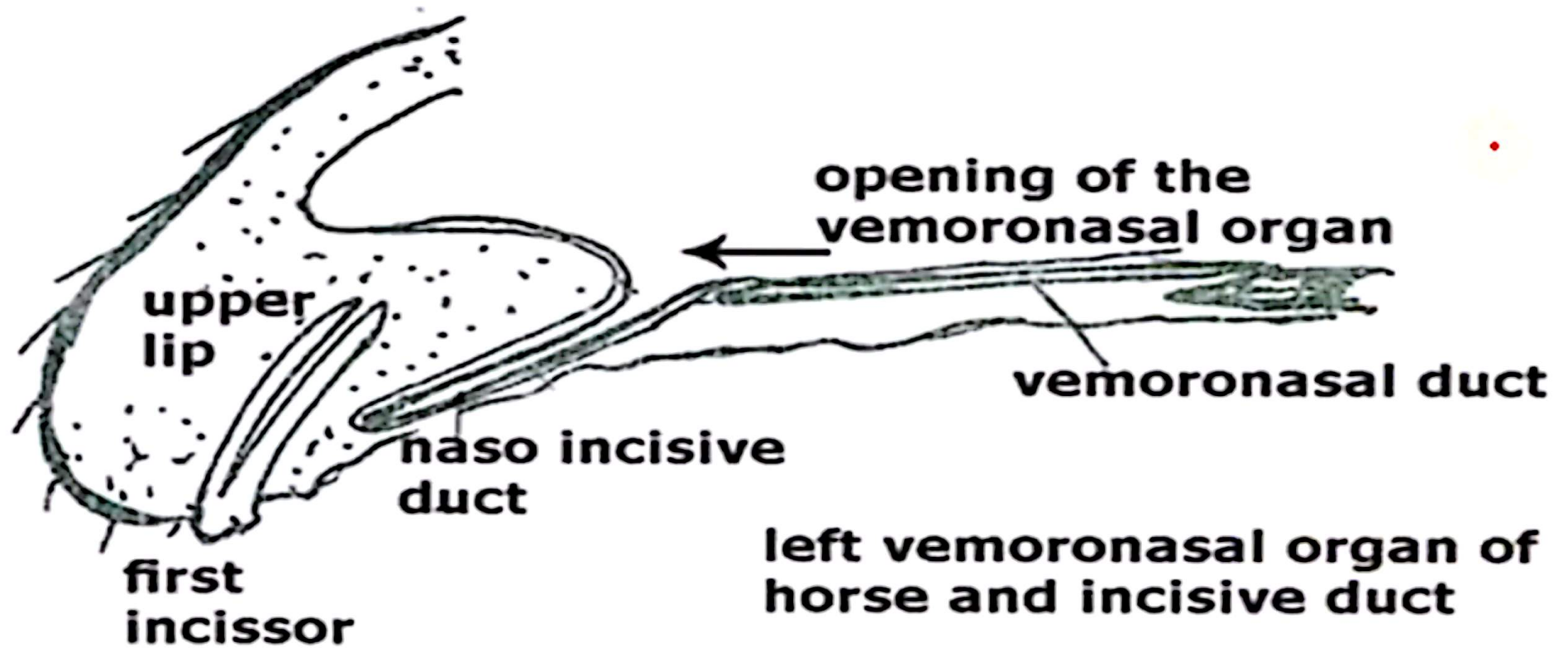
it is paired ducts present in the floor of nasal cavity and contains respiratory and olfactory elements . The duct was supported by cartilaginous sleeves

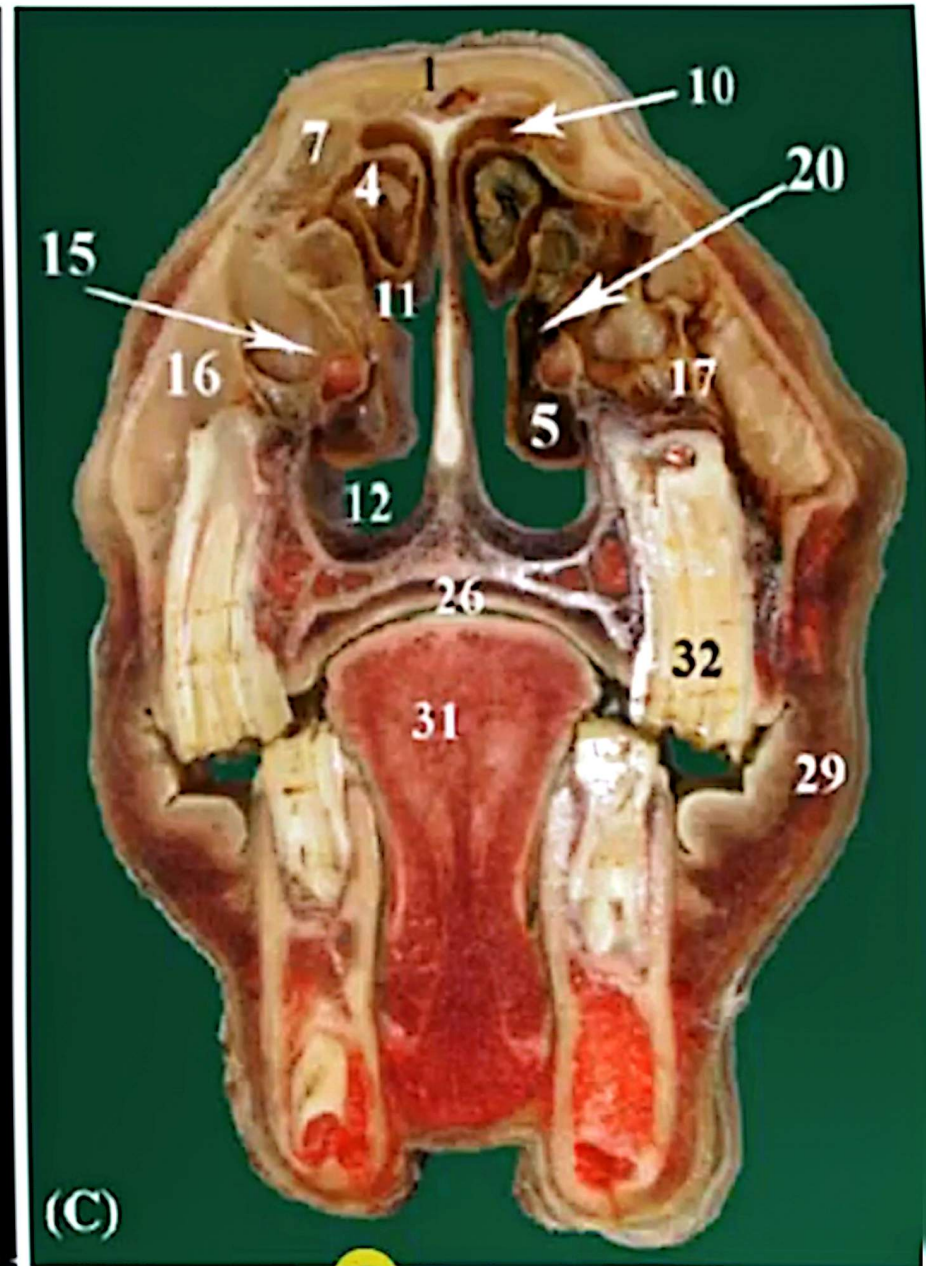
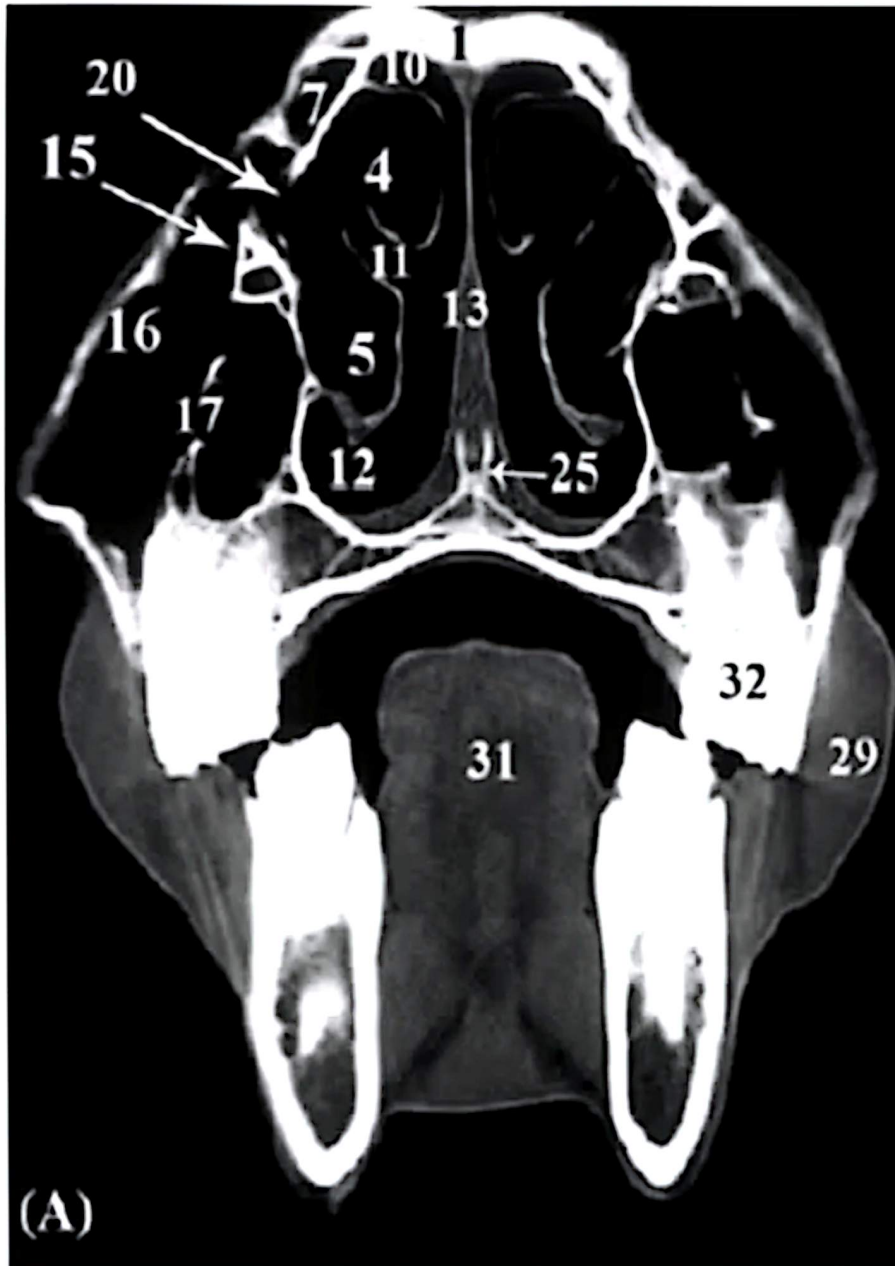
it extended from the incisive duct at level of canine teeth to 2nd or 4th cheek teeth. special olfactory functions

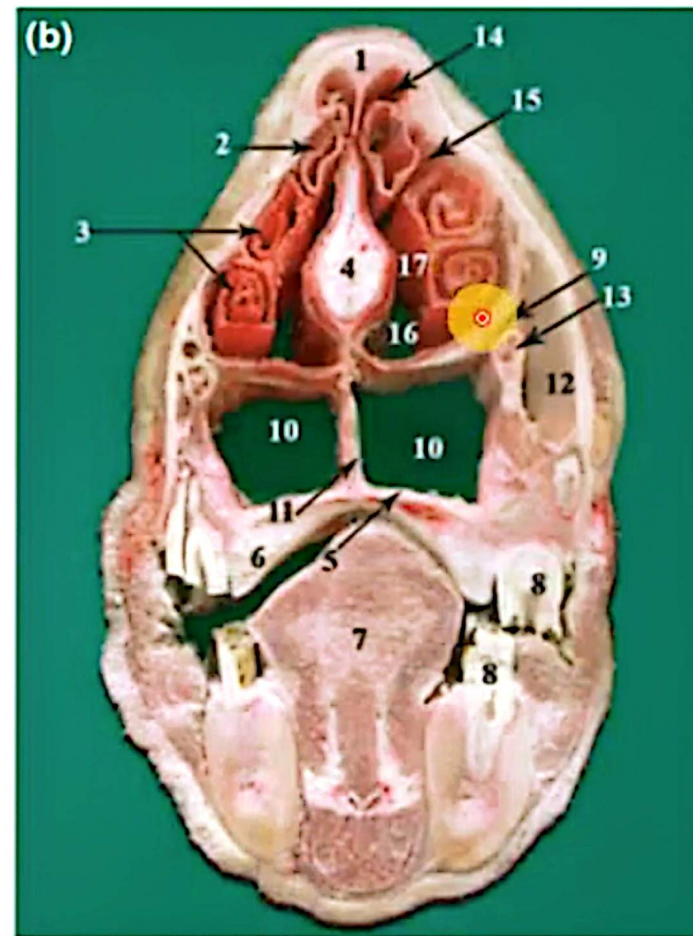
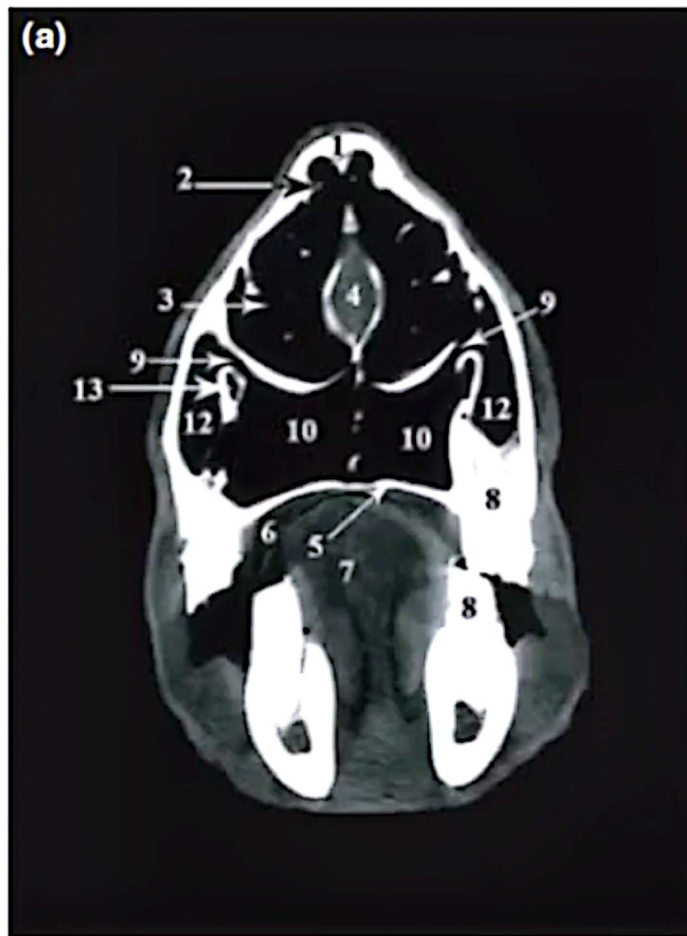
Incisive duct: (nasopalatine duct):

It is a paired tube in the floor of the nasal cavity which is directed rostroventrally and connect the nasal cavity with the oral cavity (It is nasal opening in ventral meatus at the level of canine tooth. The oral opening of the duct on the incisive papilla.

In horse the duct does not open into oral cavity... -







puted tomography scans on the left side and (b) cross-sectional image on the right side at the level of the 3rd premolar tooth. 1. Dorsal conchal sinus. 2. Ventral nasal concha. 3. Nasal septum. 4. Palatine bone. 5. Soft palate. 6. Tongue. 7. 3rd right premolar tooth. 8. Maxillopalatine opening. 9. Palatine sinus. 10. Interpalatine septum. 11. Maxillary sinus. 12. Infraorbital canal. 13. Middle nasal meatus. 14. Ventral nasal meatus. 15. Common nasal meatus. 16. Dorsal conchal sinus. 17. Ventral nasal concha.

8.27 x 11.06 in

III

and lined by m.m and continuous with the nasal cavity

1- maxillary 2- frontal 3- palatine 4- sphenoid

5- lacrimal (sphenopalatin)

Horse: maxillary – frontal- sphenopalatin

Ox1...5 Dog . M (recess) frontal

	Horse	ox	Sheep	Camel	Dog
Maxillary	+	+	+	+	Recess
Frontal	+	+	+	+	+
Lacrimal	-	+	+	+	-
Palatine	-	+	+	+	-
Sphenoid	-	+	-	-	-
sphencpalatine	+	-	-	-	-

